



STA 4222: Sample Survey Design
 Department of Mathematics & Statistics
 College of Arts & Sciences, University of South Florida

COURSE SYLLABUS

Instructor:	Xiankui Yang, M.A.	Term:	Spring 2026
Office:	CMC 015	Class Meeting Days:	T & R
Phone:	813-277-8988	Class Meeting Hours:	2:00 - 3:15 pm
E-Mail:	xiankuiyang@usf.edu	Class Location:	CMC 013
Office Hours:	T & R 1:00 – 2:00 pm	Office Hour Location:	CMC 015

I. Course Prerequisites

STA 2023 Introduction to Statistics I and STA 3024 Introduction to Statistics II

II. Course Objectives

This course is designed to teach students the commonly used sample survey design methods including simple random sampling, stratified sampling, systematic sampling, multi-stage cluster sampling, and two-phase sampling. The course also covers topics such as bias, non-sampling errors, and complex surveys.

III. Student Learning Outcomes:

Upon successful completion of this course, students will be able to:

- Understanding basic elements and general issues in sample survey design
- Distinguish between different sampling strategies and their comparisons under different conditions
- Choose and apply appropriate sampling strategies considering specific survey objectives and practical constraints
- Apply direct and/or model-assisted methods for estimation and variance quantification
- Use statistical software to implement the design and analysis for sample surveys

IV. Required Textbooks and Materials

Textbook: *Sampling: Design and Analysis* by Sharon L. Lohr, 2010, Cengage learning, 2nd Edition.

Software: R, free for download at <http://cran.us.r-project.org/> or SAS (available through Apps.usf.edu)

V. Topics covered

- **Introduction** to basic elements in sample surveys (requirements of good samples, selection bias, measurement errors, questionnaire design, sampling and non-sampling errors)
- **Simple random sampling** (sampling weights, mean and variance estimation/inference, sample design/sample size determination)
- **Systematic sampling** (implementation procedure, estimation)
- **Stratified sampling** (forming strata, estimation, sample allocation, quota sampling)
- **Ratio and regression estimation** (auxiliary variables, estimation in domains, post-stratification, design vs. model-based inference)
- **Cluster sampling with equal probabilities** (design and analysis for one-stage and two-stage cluster samples)

- **Cluster sampling with unequal probabilities** (unequal clusters, probability proportional to size sampling)
- **Nonresponse** (effects and remedies, callbacks and two-phase sampling, weighting methods, imputation)
- **Complex surveys** and related issues (sampling weights, data presentation, variance estimation, etc.)

VI. Grading Policy:

Assessment	Percent of Final Grade
Homework	30%
Midterm Exam I	20%
Midterm Exam II	20%
Final Exam	30%
	100%

97 - 100	A+	87 - 89	B+	77 - 79	C+	67 - 69	D+
94 - 96	A	84 - 86	B	74 - 76	C	64 - 66	D
90 - 93	A-	80 - 83	B-	70 - 73	C-	60 - 63	D-
						0 - 59	F

VII. Course Policies: Course Work and Grades

Homework Policy:

- Homework problems offer practices for applying lecture materials; it is an important and most effective way for mastering the course material.
- There will be **six** homework assignments. All homework will be posted on Canvas with their due dates written on the assignment sheets.
- All homework should be handed in paper copies in class. No late homework or homework sent via email will be accepted. The lowest score among the five graded homework will be dropped from the final grade calculation.
- You are encouraged to work in groups on homework problems; however, you must write your own answers. Copying others' answers is not a good way to learn.

Exam Policy:

- There will be **two midterm exams** and **one final exam**. All the exams will be closed book.
- The midterm exams will be written in class. The coverage of the exam will be announced a week before the scheduled exam date. The final exam will be cumulative.
- Makeup exams will be given only if students contact the instructor and get approval **prior** to the scheduled exam. In the event of an unforeseen emergency that prevents the student from taking the exam as scheduled, the student must provide documentation to the instructor before a makeup exam can be arranged. However, **different** questions will be used for makeup exams.

Grades of "Incomplete":

The current university policy concerning incomplete grades will be followed in this course. Incomplete grades are considered as unusual exceptions. They are given only in situations where a student is passing the course and has completed 2/3 of the course material, however was kept from completing the course

due to unexpected medical or family emergencies. Incomplete work must be finished by the end of the subsequent semester or the “I” will automatically be recorded as an “F” on your transcript.

VIII. Course Policies: Technology and Media

Email: Emails will be used to make announcements and answer class related questions. I’ll try to respond as quickly as I can during week days. However, if a student plans to ask questions or send any inquiry via emails, please make sure enough time is allowed to respond.

Canvas: Course materials including class notes, homework assignments and solutions, exam solutions, and other supporting materials will be posted on Canvas. It is the students’ responsibility to login to Canvas several times per week to check for updates.

Classroom Devices: Students can use calculators and laptops in classes. However, no tape recorders, cameras, or other video/audio & technology devices are allowed in the classroom.

IX. Course Policies: Student Expectations

Disability Access:

The University of South Florida is committed to providing reasonable accommodations for all persons with disabilities. Students with disabilities are responsible for registering with the USF Students with Disabilities Services office in order to receive special accommodations and services. Students are required to give reasonable notice (typically 5 working days) prior to requesting an accommodation. A letter from the USF Students with Disabilities Services office must accompany this request.

Attendance Policy: [USF General Attendance Policy](#)

Attendance at class meetings is essential to academic success. Full time attendance to every class is strongly recommended. First class attendance is required by the university for students who registered prior to the first day of the class.

Professionalism Policy:

Per university policy and classroom etiquette; mobile phones, iPods, *etc.* **must be silenced** during all classroom lectures. Those not heeding this rule will be asked to leave the classroom immediately so as to not disrupt the learning environment. Please arrive on time for all class meetings. Students who habitually disturb the class by talking, arriving late, *etc.*, and have been warned may suffer a reduction in their final class grade.

Academic Conduct Policy:

Academic dishonesty in any form will not be tolerated. If you are uncertain as to what constitutes academic dishonesty, please consult the University of South Florida's Student Handbook for further details. Violations of these rules will result in a record of the infraction being placed in your file and receiving a zero on the work in question AT A MINIMUM. At the instructor’s discretion, you may also receive a failing grade for the course. Confirmation of such incidents can also result in expulsion from the University

Important Notice for USF Students: In the event of an emergency, it may be necessary for USF to suspend normal operations. During this time, USF may opt to continue delivery of instruction through methods that include but are not limited to Canvas and email messaging and/or an alternate schedule. It’s the responsibility of the student to monitor the Canvas site for each class for course specific

communication, and the main USF, College, and department websites, emails, and MoBull messages for important general information.

X. Important Dates

Note that all the dates listed below are tentative and can be changed at the discretion of the professor.

Midterm Exam I:	Thu, Feb 19
Midterm Exam II:	Thu, Mar 26
Final Examination:	Tue, May 5 (12:30 – 2:30 pm)
No Class Dates:	Tue, Mar 17 (Spring break) Thu, Mar 19 (Spring break)